An Algebraic Identity and a Numerical Symmetry-Breaking Effect in a One-Parameter Family of Functions

Abdulla M. N. Al-Jabri*
Independent Researcher
Sana'a, Yemen

October 2026

Abstract

We study a one-parameter family of real functions $g(u; x_p)$ on $(0, 2\pi)$ defined by

$$g(u; x_p) = \left(\frac{u}{2\pi}\right)^5 \ln\left(\frac{2\pi}{u}\right) e^{-2\pi/(x_p u)},$$

with $x_p > 0$. The family is motivated by, but strictly independent of, the arithmetic of the second non-trivial zero of the Riemann zeta function; the specific value $x_p = 21$ is a modelling choice.

We establish four results. First, an exact algebraic identity $H_g(gY) = g(-Y'')$ relating the standard second-derivative operator acting on Y to a first-order-corrected operator $H_g = -\partial_u^2 + 2p\partial_u + (p' - p^2)$, where $p = g'/g$. Second, an exact symmetry argument showing that a matrix element of the operator $D(u) = \cos(u/2)$ between normalized states vanishes in a reflection-symmetric reference model. Third, a numerical demonstration that the asymmetry of g in $(0, 2\pi)$ breaks this vanishing: for $x_p = 21$, the specified transition operator, and the specified unweighted L^2 normalization, composite Simpson quadrature reproducibly yields $R_{13/12} \approx 0.6035155248$. Fourth, we show that g is a formal zero mode of the operator $L = -\partial_u^2 + g''/g$, which factors as $L = (-\partial_u - p)(\partial_u - p)$.

The results are either algebraic identities or numerical observations for the stated setup; Section 1.3 states the exact scope of the claims.

Keywords: algebraic gauging; Sturm-Liouville theory; symmetry-breaking; numerical integration.
MSC 2020: 34B24, 65D30, 81Q10.

Contents

1 Introduction **3**
1.1 Context . . . 3
1.2 What is specific to this kernel . . . 3
1.3 Scope of claims . . . 3

2 Setup **4**
2.1 The kernel and its logarithmic derivative . . . 4
2.2 The differential operator H_g . . . 4
2.3 States, normalization, and matrix elements . . . 4

*Independent Researcher, Sana'a, Yemen. ORCID: 0009-0003-3319-3822. Email: jabri62018@gmail.com.

3 The gauging identity **5**

4 Symmetry argument and its failure **5**
4.1 Parity of the states and of the transition operator 5
4.2 Vanishing in the symmetric reference model 6
4.3 The asymmetric weight and numerical symmetry-breaking 6

5 The operator L and its formal zero mode **7**
5.1 The operator L 7
5.2 The formal zero mode 7

6 Asymptotics of g and $V_{\mathbf{eff}}$ **7**

7 Discussion and open questions **8**
7.1 Summary 8
7.2 Open questions 8

A Reproducibility: Python code **9**

1 Introduction

1.1 Context

A companion project of exploratory documents [1, 2] investigates elementary functions on $(0, 2\pi)$ constructed from the ordinates γ_k of the non-trivial zeros of the Riemann zeta function. The present note isolates the mathematical content of two of those documents and states it in a self-contained form suitable for external review.

The central object is the family (1) below, parameterized by $x_p > 0$. The choice $x_p = 21 = \lfloor \gamma_2 \rfloor$, where $\gamma_2 \approx 21.022$ is the second positive non-trivial zero of the Riemann zeta function, is used throughout for numerical illustrations. It is a *modelling choice*; no arithmetic property of γ_2 enters any proof.

1.2 What is specific to this kernel

The individual techniques used here are standard in Sturm–Liouville theory and in supersymmetric quantum mechanics [5, 6]. The contribution of the present note is their application to the specific kernel (1), and the following observations:

1. **Structural asymmetry.** Unlike exactly solvable problems whose potentials are reflection-symmetric about the midpoint of the interval, the effective potential $V_{\text{eff}} = g''/g$ associated with (1) has a positive fourth-order pole at one endpoint and a negative first-order pole at the other (Section 6). The reflection symmetry that would force the vanishing of certain matrix elements (Proposition 3) is therefore broken.

2. **Quantitative fingerprint.** For the specific kernel, parameter value, transition operator, and normalization considered here, the asymmetry produces the numerical value $R_{13/12} \approx 0.6035155248$ (Observation 4). This is a numerical observation for the stated setup; no universality is claimed.

3. **Formal zero mode.** The kernel g itself is a formal zero mode of the operator L, giving a supersymmetric factorization (Proposition 5).

1.3 Scope of claims

What is proven. Proposition 1 (gauging identity) is proven for any C^2 function Y and any $g \in C^2(0, 2\pi)$ with $g > 0$. Proposition 3 (vanishing in the reference model) is proven for any continuous, positive, reflection-symmetric weight h with finite normalization integrals. Proposition 5 (formal zero mode and factorization) is proven as an algebraic identity. The endpoint asymptotics of Section 6 are derived in full.

What is observed numerically. For the specified kernel, parameter value, transition operator, and normalization, composite Simpson quadrature reproducibly gives $R_{13/12} \approx 0.6035155248$ (Observation 4), in agreement with an independent adaptive-quadrature reference.

What is not claimed. No proof of, or contribution to, the Riemann Hypothesis. No spectral theorem for H_g or L on a self-adjoint realization. No cosmological or physical interpretation. No universality of $R_{13/12}$ across alternative normalizations or transition operators. The choice $x_p = 21$ carries no arithmetic necessity.

2 Setup

2.1 The kernel and its logarithmic derivative

Throughout, u ranges over $(0, 2\pi)$ and $x_p > 0$. We write

$$g(u) := \left(\frac{u}{2\pi}\right)^5 \ln\left(\frac{2\pi}{u}\right) e^{-2\pi/(x_p u)}, \tag{1}$$

suppressing the dependence on x_p when no confusion can arise. The logarithmic derivative is

$$p(u) := \frac{g'(u)}{g(u)} = \frac{5}{u} - \frac{1}{u\ln(2\pi/u)} + \frac{2\pi}{x_p u^2}, \tag{2}$$

and its derivative is

$$p'(u) = -\frac{5}{u^2} + \frac{\ln(2\pi/u) - 1}{u^2 \ln^2(2\pi/u)} - \frac{4\pi}{x_p u^3}. \tag{3}$$

The function g is strictly positive on $(0, 2\pi)$ and vanishes at both endpoints; its asymptotics are analyzed in Section 6. We record for later use that g decays faster than any power of u as $u \to 0^+$, but near $u = 2\pi$ it vanishes only linearly in $t = 2\pi - u$.

2.2 The differential operator H_g

Let H_g denote the second-order differential operator

$$H_g := -\partial_u^2 + 2p(u)\,\partial_u + p'(u) - p(u)^2, \tag{4}$$

acting on C^2 functions. This operator is not self-adjoint with respect to the standard inner product on $L^2((0, 2\pi))$; it carries a first-order correction $2p\,\partial_u$, which is the origin of the gauging identity.

2.3 States, normalization, and matrix elements

For $n \in \mathbb{N}_{\geq 1}$ let $Y_n(u) := \sin(nu/2)$. Assume the weight g is such that

$$N_n := \int_0^{2\pi} g(u)^2 Y_n(u)^2\, du \in (0, \infty), \tag{5}$$

which holds for g as in (1). Define the normalized states

$$\Psi_n(u) := \frac{g(u)Y_n(u)}{\sqrt{N_n}}. \tag{6}$$

For the transition operator $D(u) := \cos(u/2)$, define

$$\mathcal{A}_{mn} := \int_0^{2\pi} \Psi_m(u) D(u) \Psi_n(u)\, du, \tag{7}$$

and the dimensionless ratio

$$R_{13/12} := \frac{|\mathcal{A}_{13}|^2}{|\mathcal{A}_{12}|^2}. \tag{8}$$

The *reference model* replaces g by a weight $h \in C^0(0, 2\pi)$ satisfying $h(2\pi - u) = h(u)$ and $h > 0$ on $(0, 2\pi)$, with normalization integrals

$$0 < \tilde{N}_n := \int_0^{2\pi} h(u)^2 Y_n(u)^2\, du < \infty, \qquad n = 1, 2, 3. \tag{9}$$

3 The gauging identity

Proposition 1 (Gauging identity)**.** *Let $g \in C^2(0, 2\pi)$ with $g(u) > 0$ on $(0, 2\pi)$, and define $p := g'/g$. Let H_g be the operator* (4)*. Then for every $Y \in C^2(0, 2\pi)$,*

$$H_g(gY) = -g\,Y''. \tag{10}$$

Proof. We compute

$$\begin{aligned} H_g(gY) &= -\partial_u^2(gY) + 2p\,\partial_u(gY) + (p' - p^2)(gY) \\ &= -(g''Y + 2g'Y' + gY'') + 2\frac{g'}{g}(g'Y + gY') + (p' - p^2)gY. \end{aligned}$$

The $g'Y'$ terms cancel. Using $p' = g''/g - p^2$, we have $p' - p^2 = g''/g - 2p^2$, hence

$$\begin{aligned} H_g(gY) &= -g''Y - gY'' + 2p^2gY + \left(\frac{g''}{g} - 2p^2\right)gY \\ &= -g''Y - gY'' + 2p^2gY + g''Y - 2p^2gY = -gY''. \end{aligned}$$

□

Corollary 2. *Suppose $Y \in C^2(0, 2\pi)$ satisfies $-Y'' = \lambda Y$ for some $\lambda \in \mathbb{R}$. Then $\psi := gY$ satisfies $H_g\psi = \lambda\psi$ in $(0, 2\pi)$.*

Remark 1. *Proposition 1 is a special case of the standard* gauging *technique in Sturm–Liouville theory: H_g is obtained from $-\partial_u^2$ by conjugation with the multiplication operator $M_g : f \mapsto gf$. The identity is classical; the specific content of the paper lies in the choice* (1) *and the numerical consequences of Sections 4 and 5.*

Corollary 2 is a statement about formal eigenfunctions. It does not promote the algebraic relation to an equality of self-adjoint operators on a Hilbert space, nor does it guarantee coincidence of boundary conditions in any self-adjoint realization. H_g is not symmetric on $C_c^\infty(0, 2\pi)$ in the standard L^2 inner product; a genuine spectral analysis requires choosing a domain invariant under H_g and analyzing its adjoint. This is not attempted here.

4 Symmetry argument and its failure

4.1 Parity of the states and of the transition operator

Consider the reflection $\hat{P} : u \mapsto 2\pi - u$. Direct computation gives

$$Y_n(2\pi - u) = (-1)^{n+1}Y_n(u), \tag{11}$$

$$D(2\pi - u) = -D(u) = (-1)^1 D(u). \tag{12}$$

Hence

$$(Y_nDY_m)(2\pi - u) = (-1)^{n+m+3}(Y_nDY_m)(u). \tag{13}$$

In particular:

- For $n = 1$, $m = 3$: exponent 7 is odd, so Y_1DY_3 is **odd** under $\hat{P}$.
- For $n = 1$, $m = 2$: exponent 6 is even, so Y_1DY_2 is **even** under $\hat{P}$. Moreover, a direct computation gives

  $$Y_1(u)D(u)Y_2(u) = \tfrac{1}{2}\sin^2(u), \tag{14}$$

  which is non-negative on $(0, 2\pi)$ and positive except on a set of measure zero.

4.2 Vanishing in the symmetric reference model

Proposition 3 (Vanishing in the reference model)**.** *Let $h \in C^0(0, 2\pi)$ satisfy $h(2\pi - u) = h(u)$ and $h > 0$ on $(0, 2\pi)$, with (9). Define*

$$\tilde{\Psi}_n(u) := \frac{h(u)Y_n(u)}{\sqrt{\tilde{N}_n}}, \qquad n = 1, 2, 3,$$

and $\tilde{\mathcal{A}}_{mn} := \int_0^{2\pi} \tilde{\Psi}_m(u)D(u)\tilde{\Psi}_n(u)\,du$. Then

$$\tilde{\mathcal{A}}_{13} = 0, \qquad \tilde{\mathcal{A}}_{12} > 0.$$

Proof. Since h is even under $\hat{P}$, so is h^2. The integrand of $\tilde{\mathcal{A}}_{13}$ is proportional to $h^2(Y_1 D Y_3)$, an even function times an odd function, hence is odd. Its integral over $(0, 2\pi)$ vanishes.

For $\tilde{\mathcal{A}}_{12}$, by (14) the integrand is

$$\tilde{\Psi}_1(u)D(u)\tilde{\Psi}_2(u) = \frac{h(u)^2}{\sqrt{\tilde{N}_1\tilde{N}_2}} \cdot \frac{1}{2}\sin^2(u).$$

Since $h > 0$ and $\sin^2(u) > 0$ almost everywhere on $(0, 2\pi)$, the integrand is strictly positive almost everywhere. With (9), this yields $\tilde{\mathcal{A}}_{12} > 0$. □

4.3 The asymmetric weight and numerical symmetry-breaking

The weight g of (1) does *not* satisfy $g(2\pi - u) = g(u)$; the ratio $g(2\pi - u)/g(u)$ differs from one. Consequently, Proposition 3 does not apply, and the vanishing of $\mathcal{A}_{13}$ is not expected to persist.

Observation 4 (Numerical symmetry-breaking)**.** *For $x_p = 21$, $D(u) = \cos(u/2)$, and the normalization (7), composite Simpson quadrature with N subintervals gives the values in Table 1. The reference value $R^* \approx 0.6035155247873$ is computed independently by adaptive quadrature (`scipy.integrate.quad`, `limit`$= 300$); the corresponding reference matrix elements are $\mathcal{A}^*_{12} \approx 0.6220748484804$ and $\mathcal{A}^*_{13} \approx -0.4832666943981$.*

Table 1: Composite Simpson quadrature for $R_{13/12}$ at $x_p = 21$, against the independent reference $R^* \approx 0.6035155247873$. Column $|R - R^*|$ is the true error; it decays rapidly with N. Column $|R - 0.60352|$ is the difference from the rounded value; since 0.60352 is not the reference but a rounding, this column converges to $|R^* - 0.60352| \approx 4.47 \times 10^{-6}$ rather than to zero and is shown for convenience only.

N	$R_{13/12}$	$\lvert R_{13/12} - R^*\rvert$	$\lvert R_{13/12} - 0.60352\rvert$
20	0.6033930653528	1.22×10^{-4}	1.27×10^{-4}
40	0.6035132002964	2.32×10^{-6}	6.80×10^{-6}
80	0.6035154874052	3.74×10^{-8}	4.51×10^{-6}
200	0.6035155246329	1.54×10^{-10}	4.48×10^{-6}
400	0.6035155247852	2.10×10^{-12}	4.47×10^{-6}

Remark 2. *The rapid decay of $|R - R^*|$ is consistent with fourth-order accuracy of composite Simpson quadrature on smooth integrands; the present data are insufficient to fit a definite empirical order (the ratios of successive errors depend on subdominant endpoint corrections). What the table establishes is that, for the specified setup, $R_{13/12}$ converges to $R^* \approx 0.6035155248$ to at least eight significant figures by $N = 200$. The value 0.6035155248 is a reproducible numerical observation for the stated setup; no universality is claimed.*

5 The operator L and its formal zero mode

5.1 The operator L

Let

$$L := -\partial_u^2 + V_{\text{eff}}(u), \qquad V_{\text{eff}}(u) := \frac{g''(u)}{g(u)}. \tag{15}$$

The operators H_g of (4) and L of (15) are distinct differential operators constructed from the same function g. The gauging identity of Proposition 1 concerns H_g; the formal zero-mode identity and factorization for L are established separately below. No claim of equivalence between the two operators is made here.

5.2 The formal zero mode

Proposition 5 (Formal zero mode)**.** *For $g \in C^2(0, 2\pi)$ with $g > 0$ on $(0, 2\pi)$ and $V_{\text{eff}} := g''/g$, the function g satisfies*

$$Lg = 0 \quad \text{on } (0, 2\pi). \tag{16}$$

Moreover, with $p := g'/g$, the operator L factors formally as

$$L = (-\partial_u - p)(\partial_u - p). \tag{17}$$

Proof. The identity (16) is a direct computation: $Lg = -g'' + (g''/g)g = 0$.

For (17), apply both sides to $f \in C^2(0, 2\pi)$:

$$\begin{aligned}(-\partial_u - p)(\partial_u - p)f &= -\partial_u(\partial_u f - pf) - p(\partial_u f - pf) \\ &= -\partial_u^2 f + (p' + p^2)f.\end{aligned}$$

Since $p' + p^2 = g''/g = V_{\text{eff}}$, the right-hand side equals Lf. □

Remark 3. *Proposition 5 is an* algebraic *statement on the open interval. It does not assert that g is a norm-square-integrable eigenfunction of a self-adjoint realization of L on $L^2((0, 2\pi))$. The potential V_{eff} has a strong singularity at $u = 2\pi$ (Section 6); a careful analysis of the self-adjoint extensions of L is required before any spectral statement can be made, and is not attempted here.*

6 Asymptotics of g and V_{eff}

Near $u = 0^+$. As $u \to 0^+$,

$$g(u; x_p) = O\left(u^5 \ln\left(\frac{2\pi}{u}\right) e^{-2\pi/(x_p u)}\right).$$

Since the exponential dominates any power of u and any power of $\ln(2\pi/u)$, we have $g(u) = O(u^N)$ as $u \to 0^+$ for every $N \in \mathbb{N}$, and $g \to 0$.

The potential V_{eff} is dominated by the $2\pi/(x_p u^2)$ term in p, so $p(u) \sim 2\pi/(x_p u^2)$ and

$$V_{\text{eff}}(u) = p(u)^2 + p'(u) \sim \frac{4\pi^2}{x_p^2 u^4} \to +\infty \quad (u \to 0^+). \tag{18}$$

Near $u = 2\pi^-$. Set $t := 2\pi - u > 0$. Since $\ln(2\pi/u) \sim t/(2\pi)$,

$$g(u) = \frac{e^{-1/21}}{2\pi} \cdot t \cdot (1 + O(t)) = C\,t + O(t^2), \qquad C = \frac{e^{-1/21}}{2\pi} \text{ for } x_p = 21. \tag{19}$$

Thus g vanishes linearly as $u \to 2\pi^-$. Correspondingly,

$$V_{\text{eff}}(u) \sim -\frac{191}{42\pi(2\pi - u)} \to -\infty \quad (u \to 2\pi^-). \tag{20}$$

Remark 4. *The two endpoint behaviors have different singular orders: a positive fourth-order pole at $u = 0$, and a negative first-order pole at $u = 2\pi$. The potential is not of the standard form required for a regular Sturm–Liouville problem; both endpoints are singular. Correspondingly, g vanishes super-polynomially at $u = 0$ but only linearly at $u = 2\pi$.*

7 Discussion and open questions

7.1 Summary

The paper establishes:

1. The gauging identity $H_g(gY) = -g\,Y''$ (Proposition 1).
2. The vanishing of $\tilde{\mathcal{A}}_{13}$ in a symmetric reference model (Proposition 3), made precise by the explicit identity (14).
3. For the specified setup, the reproducible numerical value $R_{13/12} \approx 0.6035155248$ at $x_p = 21$ (Observation 4).
4. The formal zero-mode identity $Lg = 0$ and factorization $L = (-\partial_u - p)(\partial_u - p)$ (Proposition 5).

The scope of these claims, and the list of what is not established, is given in Section 1.3.

7.2 Open questions

1. *Domain and boundary conditions for L.* Determine the self-adjoint extensions of L on $L^2((0, 2\pi))$, and whether any admits g as a genuine eigenstate. The singularity structure (18)–(20) suggests that limit-circle behavior may be relevant at $u = 2\pi$.
2. *Spectral density of L.* If L admits a self-adjoint realization with discrete spectrum $\{\lambda_n\}$, determine the asymptotic behavior of λ_n as $n \to \infty$ and the properties of the associated spectral zeta function.
3. *Dependence of $R_{13/12}$ on x_p.* A systematic study of $R_{13/12}(x_p)$ across $(0, \infty)$ is left for future work; no claim about monotonicity or continuity in x_p is made here.
4. *Alternative normalizations.* The choice (7) reflects a specific weighted L^2 inner product. How does $R_{13/12}$ vary under alternative normalizations, and what is the correct interpretation?

Acknowledgments

The author thanks an anonymous reviewer for many useful remarks that substantially improved the scope, honesty, and clarity of the paper, and for independently verifying the numerical value of $R_{13/12}$.

Data availability

The Python code in Appendix A reproduces both the Simpson quadrature table (Table 1) and the reference value via adaptive quadrature.

A Reproducibility: Python code

The following Python 3 code (requiring `numpy` and `scipy`) reproduces Observation 4 and Table 1. The function `simpson` implements the composite Simpson rule; `ratio_simpson` evaluates $R_{13/12}$ at a given number of subintervals N; `reference_value` uses `scipy.integrate.quad` to compute the reference value R^*.

```
import numpy as np
from scipy.integrate import quad

def g(u, xp=21.0):
    if u <= 0 or u >= 2*np.pi:
        return 0.0
    return (u/(2*np.pi))**5 * np.log(2*np.pi/u) * np.exp(-2*np.pi/(xp*u))

def Y(n, u):
    return np.sin(n*u/2.0)

def D(u):
    return np.cos(u/2.0)

def simpson(f, a, b, N):
    if N % 2 != 0:
        N += 1
    h = (b - a) / N
    s = f(a) + f(b)
    for i in range(1, N, 2):
        s += 4 * f(a + i*h)
    for i in range(2, N, 2):
        s += 2 * f(a + i*h)
    return s * h / 3

def ratio_simpson(N, xp=21.0):
    a, b = 0.0, 2*np.pi
    N1 = simpson(lambda u: (g(u,xp)*Y(1,u))**2, a, b, N)
    N2 = simpson(lambda u: (g(u,xp)*Y(2,u))**2, a, b, N)
    N3 = simpson(lambda u: (g(u,xp)*Y(3,u))**2, a, b, N)
    A12 = simpson(lambda u:
        (g(u,xp)*Y(1,u)/np.sqrt(N1)) * D(u) * (g(u,xp)*Y(2,u)/np.sqrt(N2)),
        a, b, N)
    A13 = simpson(lambda u:
        (g(u,xp)*Y(1,u)/np.sqrt(N1)) * D(u) * (g(u,xp)*Y(3,u)/np.sqrt(N3)),
        a, b, N)
    return A12, A13, (A13/A12)**2

def reference_value(xp=21.0, limit=300):
```

```
    N1 = quad(lambda u: (g(u,xp)*Y(1,u))**2, 0, 2*np.pi, limit=limit)[0]
    N2 = quad(lambda u: (g(u,xp)*Y(2,u))**2, 0, 2*np.pi, limit=limit)[0]
    N3 = quad(lambda u: (g(u,xp)*Y(3,u))**2, 0, 2*np.pi, limit=limit)[0]
    A12 = quad(lambda u:
        (g(u,xp)*Y(1,u)/np.sqrt(N1)) * D(u) * (g(u,xp)*Y(2,u)/np.sqrt(N2)),
        0, 2*np.pi, limit=limit)[0]
    A13 = quad(lambda u:
        (g(u,xp)*Y(1,u)/np.sqrt(N1)) * D(u) * (g(u,xp)*Y(3,u)/np.sqrt(N3)),
        0, 2*np.pi, limit=limit)[0]
    return A12, A13, (A13/A12)**2

if __name__ == '__main__':
    xp = 21.0
    A12r, A13r, Rr = reference_value(xp)
    print(f"Reference (adaptive quad):")
    print(f"  A12 = {A12r:.13f}")
    print(f"  A13 = {A13r:.13f}")
    print(f"  R   = {Rr:.13f}")
    print()
    print(f"Composite Simpson convergence at xp = {xp}:")
    print(f"{'N':>6} | {'R':>16} | {'|R-R*|':>12} | {'|R-0.60352|':>12}")
    print("-" * 60)
    for N in [20, 40, 80, 200, 400]:
        A12, A13, R = ratio_simpson(N, xp)
        print(f"{N:>6} | {R:.13f} | {abs(R-Rr):.4e}   | {abs(R-0.60352):.4e}")
```

Expected output:

```
Reference (adaptive quad):
  A12 = 0.6220748484804
  A13 = -0.4832666943981
  R   = 0.6035155247873

Composite Simpson convergence at xp = 21.0:
     N |                R |       |R-R*| | |R-0.60352|
------------------------------------------------------------
    20 | 0.6033930653528 | 1.2246e-04   | 1.2693e-04
    40 | 0.6035132002964 | 2.3245e-06   | 6.7997e-06
    80 | 0.6035154874052 | 3.7382e-08   | 4.5126e-06
   200 | 0.6035155246329 | 1.5441e-10   | 4.4753e-06
   400 | 0.6035155247852 | 2.1000e-12   | 4.4750e-06
```

Values in the last one or two digits may vary across platforms due to floating-point accumulation in the adaptive quadrature routine and in the composite Simpson sums. The values above are for reference and correspond to `numpy` 1.26 and `scipy` 1.13 on a standard x86-64 platform.

The rounded value 0.60352 appearing in the last column of Table 1 is a rounding, not an exact value; the difference $|R^* - 0.60352| \approx 4.47 \times 10^{-6}$ is therefore a floor, not a measure of quadrature error.

References

[1] A. M. N. Al-Jabri, *The Numerical Framework: A Damping Model, the Riemann Zeta Zeros, and the Proposed Integer Cutoff $x_p = 21$*, Al-Hindara Working Paper No. 1, Zenodo (2026).

DOI: 10.5281/zenodo.23540283.

[2] A. M. N. Al-Jabri, *The Tree and the Shadow: Geometric Derivation of the Reciprocal Silver Ratio and an Exploratory Statistical Analysis of Tree Crown Ratios*, Al-Hindara Working Paper No. 2, Zenodo (2026). DOI: 10.5281/zenodo.23540796.

[3] B. Riemann, *Über die Anzahl der Primzahlen unter einer gegebenen Größe*, Monatsber. Königl. Preuß. Akad. Wiss. Berlin (1859), 671–680.

[4] A. M. Odlyzko, *Tables of zeros of the Riemann zeta function*, https://www.dtc.umn.edu/~odlyzko/zeta_tables/zeros1 (accessed October 2026).

[5] G. Teschl, *Mathematical Methods in Quantum Mechanics: With Applications to Schrödinger Operators*, 2nd ed., American Mathematical Society, Providence, RI, 2014.

[6] F. Cooper, A. Khare, and U. Sukhatme, *Supersymmetry and Quantum Mechanics*, Phys. Rep. **251** (1995), 267–385.